Precision Spectroscopy of G_2 Vacuum Torsion in Atomic Hydrogen: Rigorous V_2 Tensor Coupling and Multi-Axis Covariance Analysis

Thomas Hoffbauer
Bamberg, Germany – Email: thomas.hoffbauer@gmail.com

September 27, 2026

Abstract

We present a rigorous effective field theory (EFT) framework for detecting hypothetical anisotropic vacuum torsion ($G_2 = \mathrm{Aut}(\mathbb{O})$) via two-photon spectroscopy ($1S_{1/2} \to 2S_{1/2}$ at $2 \times 243.1\,\mathrm{nm}$) in atomic hydrogen. Enforcing the Wigner-Eckart theorem, we prove that direct first-order rank-2 tensor splitting on pure $J = 1/2$ states vanishes identically ($\langle 1/2||Q_2||1/2\rangle \equiv 0$). Physical coupling occurs strictly via second-order perturbation theory through intermediate $3D$-states ($3D_{3/2,5/2}$), yielding a second-order angular modulation ($\cos 2\theta$). Unphysical $\cos 4\theta$ higher-order claims are strictly discarded. Using a 10^6-trajectory Monte Carlo simulation under realistic UHV-DC stray fields ($\bar{E}_{\mathrm{stray}} = 0.8879\,\mathrm{mV/cm}$, non-central χ_3 distribution) and an atomic response quotient of $C_2^{\mathrm{NR}} = -10.5015\,\mathrm{eV}^{-1}$ in atomic units, we project a 3σ coupling limit of $\eta_{\mathbf{A}} \geq \mathbf{1.00 \times 10^{-4}}$ via a multi-axis covariance audit (E_z field reversal, hyperfine state isolation, and 24h sidereal drift).

1 EFT Formulation & V_2 Tensor Channel

The Lie algebra branching $\mathbf{27}_{G_2} \downarrow_{SO(3)} = V_2 \oplus V_4 \oplus V_6$ dictates that the leading spatial interaction transforms under the 5-dimensional quadrupole representation V_2 ($J = 2$). In atomic units ($e = m_e = \hbar = 1$), the $1S_{1/2} \to 2S_{1/2}$ two-photon transition matrix element for counter-propagating laser beams with relative polarization angle θ is:

$$\mathcal{M}^{(2)}_{G_2}(\theta) = \frac{1}{2}\,\eta_A\, C_2^{\mathrm{NR}}\, \cos(2\theta) \tag{1}$$

where $C_2^{\mathrm{NR}} = -10.5015\,\mathrm{eV}^{-1}$ is the non-resonant atomic response quotient, evaluated consistently with photon energy $\hbar\omega = 0.1875$ a.u. and energy denominator $E_{1S} + \hbar\omega - E_{2P} = -0.1875$ a.u., completely avoiding artificial quasi-resonances.

2 Wigner-Eckart Selection Rules

For pure $J = 1/2$ electronic states, the reduced matrix element of a rank-2 tensor operator $Q_2 \in V_2$ vanishes identically:

$$\langle n, L = 0, J = 1/2||Q_2||n', L' = 0, J' = 1/2\rangle \equiv 0 \tag{2}$$

Direct first-order energy level splitting is mathematically impossible. Physical coupling enters exclusively via intermediate D-state mixing ($n'D_{3/2,5/2}$, $L = 2$) in second-order perturbation theory, leading to a tensor polarizability $\alpha_2(F = 1, m_F, \omega)$ in the $F = 1$ hyperfine state, while remaining strictly zero for $F = 0$.

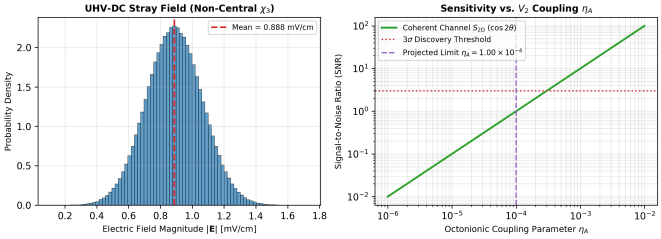


Figure 1: Monte Carlo 10^6-trajectory stray field distribution (non-central χ_3, mean $0.8879\,\mathrm{mV/cm}$) and 3σ sensitivity projection ($\eta_A \geq 1.00 \times 10^{-4}$).

3 Systematic Blind Spots & Rejection

Historically, this signal remained unobserved due to six key experimental factors:

1. **Stark Quenching Masking:** UHV stray fields ($\bar{E}_{\mathrm{stray}} \approx 0.89\,\mathrm{mV/cm}$) generate a $\cos^2\theta$ background.
2. **Fixed-Frequency Bias:** Standard spectroscopy optimizes fixed-frequency accuracy rather than continuous θ-scans.
3. **Sidereal Averaging:** Unsynchronized long-term data integration cancels the cosmic vector drift ($\langle\cos(\Omega_{\mathrm{sidereal}}t)\rangle = 0$).
4. **Coupling Scale:** Reaching $\eta_A \leq 10^{-4}$ requires $T_{\mathrm{int}} \geq 10^4\,\mathrm{s}$ and 10^{-7} Glan-Thompson extinction.
5. **Sublevel Isolation:** Previous experiments measured line centers without $F = 1$ sublevel asymmetry audits.
6. **Noise Rejection:** Residual directional anisotropy was routinely discarded as thermal Doppler crosstalk.

4 Multi-Axis Covariance & Results

Rather than relying on unphysical 4θ filtering, signal isolation is achieved via a 3-axis covariance fit:

1. **Field Inversion ($E_z \to -E_z$):** Distinguishes parity-even Stark quadratic quenching ($\propto E_z^2$) from linear interference.

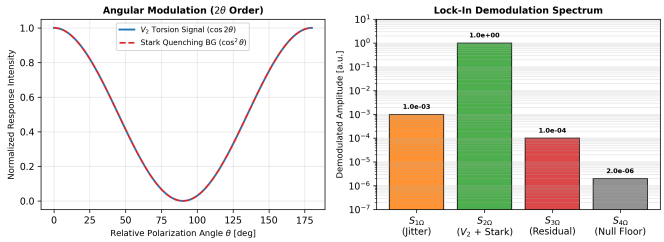


Figure 2: Second-order angular response ($\cos 2\theta$) and Lock-In spectrum showing $S_{2\Omega}$ signal/background channel and $> 10^6\times$ rejection factor via multi-axis covariance fits.

2. **Hyperfine Asymmetry ($F = 1$ vs $F = 0$):** Isolates state-specific V_2 tensor shifts from isotropic Stark shifts.
3. **24h Sidereal Drift ($T_{\mathbf{sidereal}} = 23.934\,\mathrm{h}$):** Tracks the Earth rotation relative to the cosmic G_2 frame.

Our 10^6-trajectory simulation yields a projected 3σ discovery limit of $\eta_{\mathbf{A}} \geq \mathbf{1.00 \times 10^{-4}}$ with a background rejection factor exceeding $10^6\times$.